Experiments on packet capture

tool : Wireshark.

Exp. no: 5

Date : 01/08/26

Aim:

Experiments on packet capture tool: Wireshark.

packet Sniffer

→ Sniffs messages being sent / received from / by your computer

→ passive program
- never sends packets itself
- no packets addressed to it
- receives a copy of all packets (sent / received)

packet Sniffer Structure Diagnostic Tools

→ Tcpdump
- Eg: tcpdump -ena host 10.129.41.2 -w exe3.out

→ Wireshark
- Wire-shark -r exe3.out.

Theory:

→ Wireshark is a network packet analyzer used to capture and inspect network traffic in real time.

→ It helps in troubleshooting network issues, analyzing protocols, monitoring traffic, and learning network communication.

→ It captures packets, decodes protocols, applies filters, displays statistics, and analyzes packet flow.

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features :-

→ captures live network traffic.
→ Supports protocols decoding
→ provides capture and display filters.
→ Displays packet details and statistics
→ Generates flow graphs.
→ Saves captured packets for later analysis.

Wireshark Interface

1.) packet List - Displays all captured packets.
2.) packet Details - Shows protocol information of the selected packet.
3.) Packet Bytes - Displays packet data in hexadecimal format.

Common Filters

→ tcp - Displays Tcp packets.
→ Udp - Displays UDP packets.
→ arp - Displays ARP packets.
→ dns - Displays DNS packets
→ http - Displays HTTP packets
→ icmp - Displays ICMP packets
→ dhcp - Displays DHCP packets

procedure

1.) Open Wireshark.

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2.) Select the Ethernet / Local Area Connection interface.

3.) Go to Capture → options

4.) Set Stop capture after 100 packets.

5.) Click Start to begin packet capture

6.) Apply the required filter (TCP, UDP, ARP, DNS, HTTP, ICMP, DHCP).

7.) Inspect the captured packets

8.) For TCP and DNS, view statistics → Flow Graph

9.) Save the captured packets.

TCP and UDP packets

→ Applied the tcp and udp display filters in Wireshark to view only TCP and UDP packets.

→ Observed communication between the source and destination using IP addresses and port numbers.

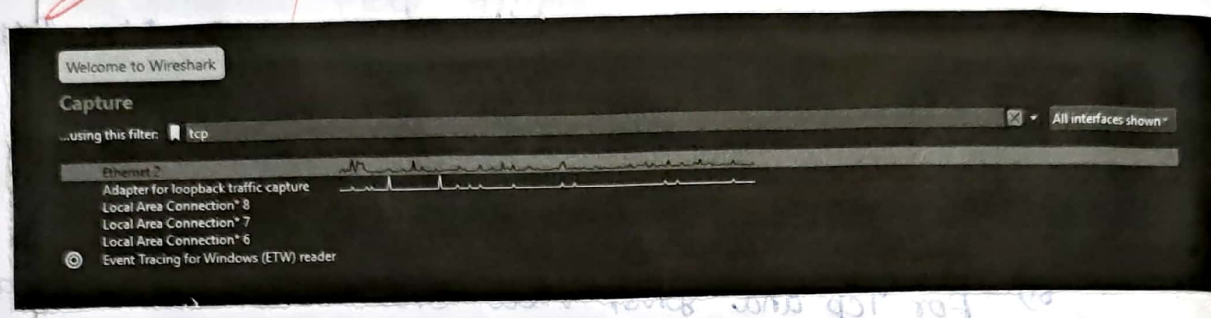
→ Tcp packets showed reliable communication with connection establishment and acknowledgements, while UDP packets showed fast, connectionless. communication.

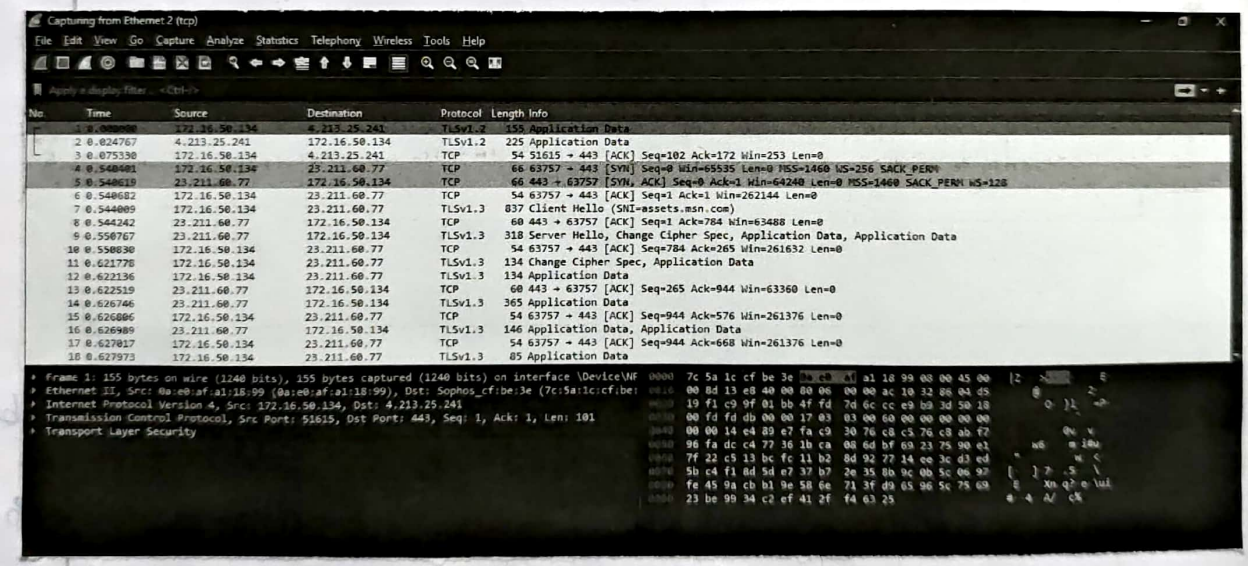
→ Examined packet details such as source / destination addresses, ports, protocol headers and payload.

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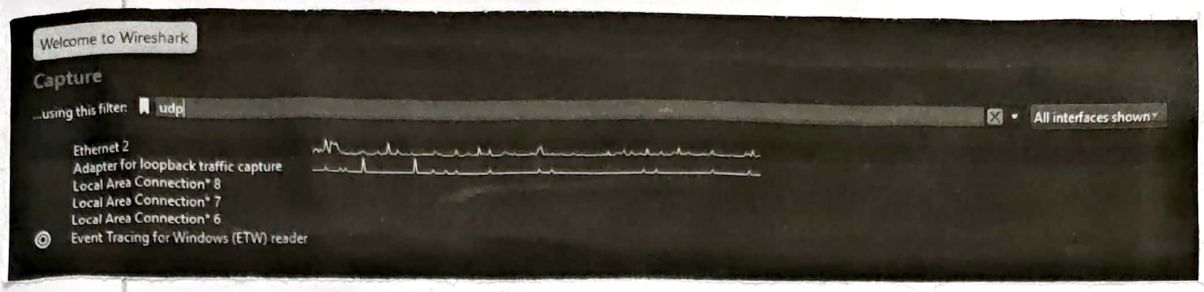
→ Viewed the communication sequence using Statistics → Flow Graph.

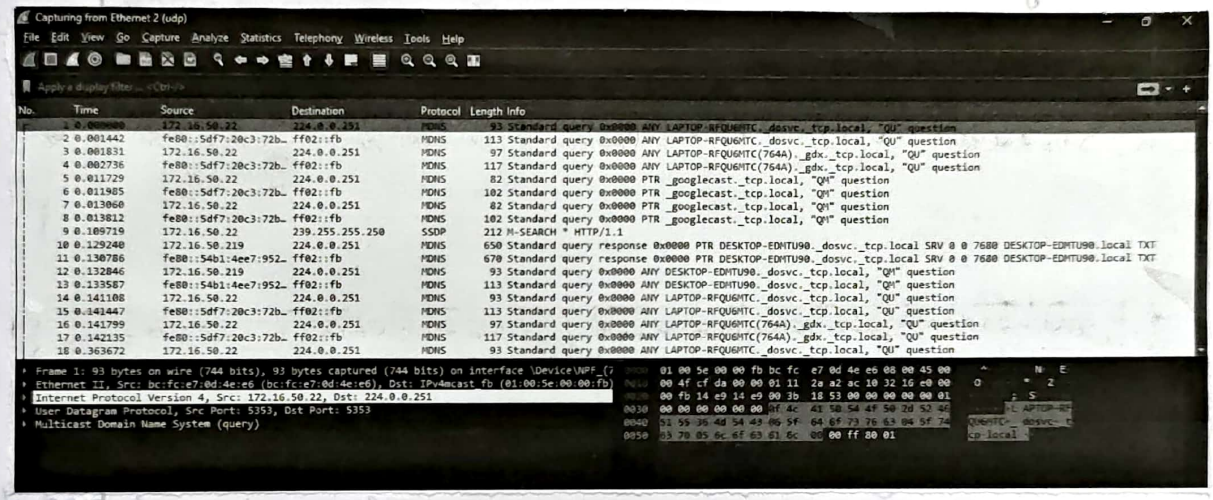
→ Successfully analyzed and saved the capture Tcp / UDP packets.





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Result :

Successfully captured, filtered, and analyzed different network packets using Wireshark. The tcp and udp ~~helps~~ packets helped understand the communication between network devices and network protocols.

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